

- LSST at Rubin Observatory
 - Billions of galaxies with many transients
 - Euclid: high resolution space imaging, produces images of galaxies and their shapes
- Problems with surveys that are essentially “too good” → Need a good way to analyze this data
 - Loss of atmospheric blurring helps with resolution and galactic analysis
- Model Rubin and Euclid galaxies together: better source detection in crowded images
 - Cosmology: can get much better joint analysis on transients for galaxies and supernovae
 - Derived Data Products (DDPs): Combine both Euclid and Rubin photometry data
 - Model all the data together, rather than enforcing design constraints - avoids assumption that any data source is the ground truth
- Need to find machine-learning priors to stably fit models to determine complex galaxy shapes (non-parametric morphology models)
 - **Question: How can we verify the data that the model is trained on, given that we’re still deriving so much data from the 2 telescope sources?**
- Scarlet2: Fully differentiable, non-parametric galaxy morphology model
 - Proximal operators help with fitting galactic models
 - Diffusion model with strong understanding of galactic shapes, helps with realistic galaxy shapes with degeneracy based on individual pixel analysis
- Does not rely on reference images: can deblend the host on its own to get detailed posteriors on several different parameters
 - Helps with overfitting by having defensible priors
- Build up samples up supernovae that can help check the accuracy of the pipeline
 - Helps with gravitationally lensed supernova imaged through multiple lenses
- Accurate light curve and time delay recovery of 0.6”!
- **Question: What do each of the different light curves tell you about the supernova, and how do you fairly combine the data analyzed from each curve?**
- Shows examples including massive black hole interactions that low-mass black holes cannot have, example: runaway blackholes
- Debris from stars falling into black holes can cause flares that last almost half a year
- Black hole disks can tear and precess! Can be detected through rotating structures in the disk and recreated in simulations to find necessary conditions